

Math 720 Notes - Spring 2026

0 Category Theory

Definition 0.1. A **category** $\mathbf{C}$ consists of:

1. a collection of **objects**, $\text{ob}(\mathbf{C})$
2. for every two objects A and B of $\mathbf{C}$, a collection $\text{Hom}_{\mathbf{C}}(A, B)$ of **morphisms**, satisfying the following properties:
 - One can compose morphisms: two morphisms $f \in \text{Hom}_{\mathbf{C}}(A, B)$ and $g \in \text{Hom}_{\mathbf{C}}(B, C)$ determine a morphism $gf \in \text{Hom}_{\mathbf{C}}(A, C)$. That is, for every triple of objects A, B and C , there is a function

$$\text{Hom}_{\mathbf{C}}(A, B) \times \text{Hom}_{\mathbf{C}}(B, C) \rightarrow \text{Hom}_{\mathbf{C}}(A, C),$$

and the image of the pair (f, g) is denoted gf .

- This ‘composition law’ is associative: if $f \in \text{Hom}_{\mathbf{C}}(A, B)$, $g \in \text{Hom}_{\mathbf{C}}(B, C)$, and $h \in \text{Hom}_{\mathbf{C}}(C, D)$, then

$$(hg)f = h(gf).$$

- For every object A of $\mathbf{C}$, there exists a morphism $1_A \in \text{Hom}_{\mathbf{C}}(A, A)$, the ‘identity’ on A , which is an identity with respect to composition: that is, for all $f \in \text{Hom}_{\mathbf{C}}(A, B)$ and $g \in \text{Hom}_{\mathbf{C}}(B, A)$ we have

$$f1_A = f, 1_Ag = g.$$

- The collections $\text{Hom}_{\mathbf{C}}(A, B)$ are disjoint.

Exercise 1. Identify some categories you’ve already seen in math. What are the objects? What are the morphisms? Do the morphisms satisfy the required axioms?

1 The Categories **Ring** and **Rng**

1.1 Ring Basics

Definition 1.1. We say R is a **ring** with operations $+$, $\cdot$ if

1. $(R, +)$ is an abelian group; we denote its additive identity by 0_R or 0 when the base ring is clear (i.e. $0 + a = a$ for all $a \in R$)
2. Multiplication is associative, that is, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in R$.
3. The operations $+$, $\cdot$ are distributive, that is $(a + b) \cdot c = a \cdot c + b \cdot c \in R$ for all $a, b, c \in R$.
4. In the course, we will also require R to have a multiplicative identity, which we denote 1_R or 1 (i.e. $1 \cdot a = a \cdot 1 = a$ for all $a \in R$). If R satisfies the all but this last condition, we will call it a **rng** (some texts will not make this distinction).

Notation: We might write $a \cdot b$ as ab for convenience.

Remark 1.2. The category **Ring** is the category whose objects are rings with a multiplicative identity, and whose morphisms are ring homomorphisms, which we will define in the next section. The category of rings where a multiplicative identity is not required is denoted **Rng**.

Exercise 2. Let R be a ring, and $a, b \in R$. Recall that $-a$ denotes the additive inverse of a . Prove the following:

1. $0 \cdot a = 0 = a \cdot 0$
2. $(-a) \cdot b = -(a \cdot b) = a \cdot (-b)$
3. $(-a) \cdot (-b) = a \cdot b$
4. 1_R is unique.

Definition 1.3. A ring is called **commutative** if $a \cdot b = b \cdot a$ for all $a, b \in R$

Definition 1.4. Let R be a ring and $a \in R$. We say a is **invertible** (or, a is a **unit**) if there exists an element $a^{-1} \in R$ such that $aa^{-1} = a^{-1}a = 1$.

Exercise 3. Let R be a ring and $a \in R$. Prove that if a is invertible, then a^{-1} is unique.

1.2 Types of Rings: Fields, Integral Domains

Definition 1.5. Let R be a ring with $1 \neq 0$. We say R is a **division ring** if every element of R is invertible. A commutative division ring is called a **field**.

Definition 1.6. Let R be a ring, and $a \in R$. We say a is a **zero-divisor** if there exists $b \in R \setminus \{0\}$ such that either $a \cdot b = 0$ or $b \cdot a = 0$.

Exercise 4. Prove that if k is a field, then 0 is its only zero-divisor.

Exercise 5. Find an example of a ring R whose only zero-divisor is 0, but which is not a field.

Exercise 6. Find all zero-divisors of $\mathbb{Z}/6\mathbb{Z}$.

Exercise 7. Let $R^* = \{u \in R \mid u \text{ is invertible}\}$, the set of units of R . Prove that $(R^*, \cdot)$ is a group.

Definition 1.7. Let R be commutative ring. Then we say R is an **integral domain** if every nonzero element of R is a non zero-divisor. Equivalently, $ab = 0 \implies a = 0$ or $b = 0$ for all $a, b \in R$.

Exercise 8. Find three examples of rings you already know, and ensure at least one example is not commutative, and at least one has a zerodivisor other than 0.

1.3 Subrings

Definition 1.8. Let R be a ring. We say that $S \subseteq R$ is a **subring** of R if $S \subseteq R$ is an additive subgroup and S is closed under multiplication ($x, y \in S \implies xy \in S$). Since we will work in the category **Ring**, we also require that $1_R \in S$.

Exercise 9. Check that if S is a subring of R , then S is also a ring itself, and $1_S = 1_R$.

1.4 Polynomial Rings

Definition 1.9. Let R be a commutative ring. Then we denote

$$R[x] := \{a_0 + a_1x + \cdots + a_nx^n \mid n \in \mathbb{N}, a_i \in R\},$$

the set of polynomials in the indeterminant x with coefficients in R . If

$$f = a_0 + a_1x + \cdots + a_nx^n \in R$$

with $a_n \neq 0$ then we say the **degree** of f is n , and denote this $\deg(f) = n$.

Exercise 10. Check that $R[x]$ is a commutative ring with respect to “the usual” operations of addition and multiplication of polynomials.

Proposition 1.10. *Let R be an integral domain, and let $f, g \in R[x]$. Then*

1. $\deg(fg) = \deg(f) + \deg(g)$
2. $R[x]$ is an integral domain
3. If $f \in R[x]$ is a unit, then $f \in R$ and f is a unit in R .

Exercise 11. Prove the proposition above.

Exercise 12. Find an example of a ring R and $f, g \in R[x]$ such that $\deg(fg) < \deg(f) + \deg(g)$.

1.5 Ring homomorphisms

Definition 1.11. Let R and S be rings. A map $\varphi : R \rightarrow S$ is called a **ring homomorphism** if for all $a, b \in R$,

1. $\varphi(a + b) = \varphi(a) + \varphi(b)$,
2. $\varphi(ab) = \varphi(a)\varphi(b)$, and
3. $\varphi(1_R) = 1_S$ (This last condition is not required for morphisms in the category **Rng**).

Exercise 13. Prove that for any ring homomorphism $\varphi : R \rightarrow S$, we have that $\varphi(0_R) = 0_S$.

Exercise 14. Let R be a subring of a ring S , and let $a \in R$. Decide which of the following are ring homomorphisms:

1. The *inclusion map*, $\iota : R \hookrightarrow S$, sending $r \mapsto r$.
2. The projection map $\pi : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ sending $z \mapsto [z]_n$
3. The “multiplication by 2” map $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}$ sending $z \mapsto 2z$.
4. The “evaluation at 1” map: $\varphi_1 : R[x] \rightarrow R$, sending $f \mapsto f(1)$.
5. The “evaluation at a ” map: $\varphi_a : R[x] \rightarrow R$, sending $f \mapsto f(a)$.

Definition 1.12. Let $\varphi : R \rightarrow S$ be any map.

Recall that φ is **injective** if $\varphi(a) = \varphi(b) \implies a = b$ and is **surjective** if for all $s \in S$ there exists an $r \in R$ such that $\varphi(r) = s$.

We say φ is **bijective** if it is both injective and surjective.

A bijective ring homomorphism is called an **isomorphism**.

Remark 1.13. Note that a bijective ring homomorphism φ has a unique inverse φ^{-1} .

Definition 1.14. Let $\varphi : R \rightarrow S$ be a ring homomorphism. Then we define the **kernel** of φ by

$$\ker \varphi =: \{x \in R \mid \varphi(x) = 0\}$$

and the **image** of φ by

$$\operatorname{im} \varphi =: \{\varphi(x) \mid x \in R\}$$

Exercise 15. Let $\varphi : R \rightarrow S$ be a ring homomorphism. Show that $\operatorname{im} \varphi$ is a subring of S .

Proposition 1.15. Let $\varphi : R \rightarrow S$ be a ring homomorphism. The following are equivalent:

1. φ is injective
2. $\varphi(x) = 0 \implies x = 0$
3. $\ker \varphi = \{0\}$

Exercise 16. Prove the proposition above.

1.6 Ideals

Definition 1.16. Let R be a ring. We say $I \subseteq R$ is a **right (respectively left) ideal** of R if:

1. I is a subgroup of $(R, +)$.
2. For every $a \in I$ and $r \in R$, we have $ar \in I$ (respectively $ra \in I$)

When I is both a right and left ideal, we call it a **two-sided ideal**, or just an **ideal**.

Example 1.17. For any $n \in \mathbb{Z}$, the set $n\mathbb{Z} = \{nz \mid z \in \mathbb{Z}\}$ forms an ideal of $\mathbb{Z}$

Exercise 17. Let R be a ring, and let $a \in R$. Prove that the set $aR = \{ar \mid r \in R\}$ is a right ideal of R .

Is $RaR = \{ras \mid r, s \in R\}$ a two sided ideal of R ?

Exercise 18. Let R be a ring and let $I \subseteq R$ be a subset of R . Explain why in order to check that I is a left ideal, it suffices to check that for every $a, b \in I$ and $r \in R$, $r \cdot (a - b)$ is an element of I (and similarly for right, two-sided).

Exercise 19. Let $\varphi : R \rightarrow S$ be a ring homomorphism. Show that $\ker \varphi$ is a two-sided ideal of R .

Note: We will soon see that every two-sided ideal of a ring R is the kernel of some ring homomorphism.

1.7 Operations with Ideals

Notation 1.18. Let R be a ring and let I, J be ideals of R . Denote:

$$I + J := \{i + j \mid i \in I, j \in J\}$$

and

$$IJ := I \cdot J := \{i_1 j_1 + \dots + i_n j_n \mid n \in \mathbb{N}, i_k \in I, j_k \in J \text{ for all } k\}$$

Proposition 1.19. *Let I, J be ideals of a ring R . Then $I + J$ is the smallest ideal containing both I and J .*

Exercise 20. Prove the proposition above.

Exercise 21. Find an example of two ideals I, J in a ring R , such that $I \cup J$ is not an ideal.

Exercise 22. Let I, J be ideals of a ring R . Prove that IJ is an ideal of R .

Proposition 1.20. Let $\{I_\lambda\}_{\lambda \in \Lambda}$ be a family of ideals of a ring R . Then $\bigcap_{\lambda \in \Lambda} I_\lambda$ is an ideal of R .

Exercise 23. Prove the proposition above.

Exercise 24. Let $m, n \in \mathbb{Z}$, and let $I = n\mathbb{Z}$ and $J = m\mathbb{Z}$. Describe the following sets:

1. $I \cap J$
2. $I \cdot J$
3. $I + J$
4. $I \cup J$

Exercise 25. Let I, J be ideals of a ring R . Show that $I \cdot J \subseteq I \cap J$, and then give an example where $I \cdot J \neq I \cap J$.

1.8 Quotient Rings

Definition 1.21. Let R be a ring, and $I \subseteq R$ a two-sided ideal. We can define an equivalence relation on R by $a \sim b \iff a - b \in I$, and denote by $\bar{a} = \{r \in R \mid r \sim a\}$. Then we define the **quotient ring**:

$$R/I = \{\bar{r} \mid r \in R\}$$

with $+, \cdot$ inherited from R , that is, $\bar{a} + \bar{b} := \overline{a + b}$ and $\bar{a} \cdot \bar{b} := \overline{ab}$.

Note: For $r \in R$, you may see the notation $\bar{r}, [r]_I$, or $r + I$ to represent elements of R/I .

Exercise 26. Explain what it means for the operations $+, \cdot$ on R/I to be *well-defined*, and then check that they are well-defined. Further, check that R/I is a ring. Why do we need I to be a two-sided ideal?

Example 1.22. For any $n \in \mathbb{Z}$, $\mathbb{Z}/n\mathbb{Z}$ forms the quotient ring with elements $\bar{0}, \bar{1}, \dots, \overline{n-1}$.

Exercise 27. Consider the map $\pi : R \rightarrow R/I$ defined by $r \mapsto \bar{r}$. Prove that π is a surjective ring homomorphism and then compute $\ker \pi$.

1.9 Isomorphism Theorems

Theorem 1.23 (The First Isomorphism Theorem). *Let $\varphi : R \rightarrow S$ be a ring homomorphism. Then*

$$\frac{R}{\ker \varphi} \cong \operatorname{im} \varphi$$

Exercise 28. Prove this theorem by checking that φ induced a well-defined ring homomorphism $\tilde{\varphi} : R/\ker \varphi \rightarrow \operatorname{im} \varphi$, and then check that it is an isomorphism.

Theorem 1.24 (The Second Isomorphism Theorem). *Let R be a ring and let S be a subring of R , and I an ideal of R . Then $S + I$ is a subring of R and*

$$\frac{S + I}{I} \cong \frac{S}{S \cap I}$$

Exercise 29. Prove this theorem by defining a map $\varphi : S \rightarrow (S + I)/I$, and then applying the first isomorphism theorem.

Theorem 1.25 (The Third Isomorphism Theorem). *Let $I \subseteq J$ be ideals of a ring R . Then J/I is an ideal of R/I and*

$$\frac{R/I}{J/I} \cong R/J$$

Exercise 30. Prove this theorem by defining a map $\varphi : R/I \rightarrow R/J$, and then applying the first isomorphism theorem.

Theorem 1.26. *Let I be an ideal of a ring R . Then there is a one-to-one correspondence*

$$\{\text{ideals of } R \text{ that contain } I\} \longleftrightarrow \{\text{ideals of } R/I\}$$

and similarly

$$\{\text{subrings of } R \text{ that contain } I\} \longleftrightarrow \{\text{subrings of } R/I\}$$

Exercise 31. Prove the first correspondence in the theorem above (the second will be similar). Define a map (of sets) $\varphi : \{\text{ideals of } R \text{ that contain } I\} \rightarrow \{\text{ideals of } R/I\}$ by $\varphi(J) = J/I$ and a map $\psi : \{\text{ideals of } R/I\} \rightarrow \{\text{ideals of } R \text{ that contain } I\}$ by $\psi(K) = \{x \in R \mid \bar{x} \in K\}$, and show these maps are inverses (and so are both bijections of sets).

1.10 Properties of Ideals

Definition 1.27. Let R be a ring with $1 \neq 0$. Let $A \subseteq R$ be a subset. We define the (two-sided) **ideal generated by A** by

$$(A) := \bigcap_{\substack{I \supseteq A \\ I \text{ ideal}}} I.$$

Some texts will denote (A) by $\langle A \rangle$. Note that since this is an intersection of ideals, it is also an ideal. Moreover, by the way it is defined, it is the *smallest* ideal that contains A .

Proposition 1.28. *Let $A \subseteq R$ be a subset, and let R be a ring. Then the ideal generated by A is given by $(A) = \{r_1 a_1 s_1 + r_2 a_2 s_2 + \cdots + r_n a_n s_n \mid n \in \mathbb{N}, a_i \in A, r_i, s_i \in R\}$.*

Proof. First, note that the set $L = \{r_1 a_1 s_1 + \cdots + r_n a_n s_n \mid n \in \mathbb{N}, a_i \in A, r_i, s_i \in R\}$ is an ideal of R (you should think about why!).

Now $A \subseteq L$ because for any $a \in A$, $a = 1 \cdot a \cdot 1 \in L$. Thus, $(A) \subseteq L$, since L is one of our ideals in the intersection defining (A) .

Now we will show that $L \subseteq (A)$. Let $\ell = r_1 a_1 s_1 + \cdots + r_n a_n s_n \in L$, where by definition, $a_i \in A$. Now let $I \subseteq R$ be an ideal. Then $r_i a_i s_i \in I$ since I is a two sided ideal, and so $\ell \in I$ since I is also closed under addition. Thus, since I was arbitrary, we have

$$\ell \in \bigcap_{\substack{I \supseteq A \\ I \text{ ideal}}} I = (A).$$

□

Note: You could similarly define the left ideal or right ideal generated by A .

Definition 1.29. Let I be an ideal. If I is generated by a single element $a \in R$, so that $I = (\{a\})$, we say I is a **principal ideal**, which we typically denote by $I = (a)$.

Definition 1.30. If A is a finite set $A = \{a_1, \dots, a_n\}$ and $I = (A)$, then we say I is a **finitely generated ideal**, which we typically denote by $I = (a_1, \dots, a_n)$.

Definition 1.31. An ideal I such that $I \neq R$ is called a **proper ideal**.

Exercise 32. An integral domain with only principal ideals is called a **principal ideal domain** or **PID**. For each of the following rings, decide whether or not it is a PID. If not, give an example of a non-principal ideal.

1. $\mathbb{Z}$

2. $\mathbb{Z}[x]$

3. $\mathbb{Q}[x]$

4. $\mathbb{Q}[x, y]$

Example 1.32. Hilbert's Basis Theorem tells us that if k is a field, all ideals in $k[x_1, \dots, x_n]$ are finitely generated. The proof of this fact is typically covered in a commutative algebra or algebraic geometry class. Rings with this property (that all ideals are finitely generated) are called **Noetherian**.

Exercise 33. Let R be a ring, and I an ideal of R . Prove that $I = R$ if and only if I contains a unit.

Proposition 1.33. *Let R be a ring. Then R is a field if and only if its only ideals are (0) and R .*

Exercise 34. Prove the proposition above.

Corollary 1.34. *Let $\varphi : k \rightarrow S$ be a ring homomorphism, where k is a field. Then either $\varphi = 0$ (the map sending every element to 0) or φ is injective.*

Proof. Since $\ker \varphi$ is an ideal of k by Exercise 19, we see by Proposition 1.33 that $\ker \varphi = 0$, in which case φ is injective by Proposition 1.15, or $\ker \varphi = k$, in which case φ is the map that sends every element of k to 0. $\square$

Definition 1.35. We say an ideal $\mathfrak{m}$ is **maximal** if $\mathfrak{m} \neq R$ and if $\mathfrak{m} \subseteq J$ for some ideal J , then $J = R$ or $J = \mathfrak{m}$.

Proposition 1.36. *Let R be a commutative ring. Then $\mathfrak{m}$ is a maximal ideal if and only if $R/\mathfrak{m}$ is a field*

Exercise 35. Prove the proposition above. (Hint: Use Proposition 1.33).

Theorem 1.37. *Let R be a ring with $0 \neq 1$. Then every proper ideal is contained in a maximal ideal.*

We will prove this theorem in a homework problem, using Zorn's Lemma.

Exercise 36. Which ideals in $\mathbb{Z}$ are maximal?

Definition 1.38. Let R be a commutative ring with $1 \neq 0$. Then we say a proper ideal $P \in R$ is **prime** if $ab \in P \implies a \in P$ or $b \in P$.

Exercise 37. Find an ideal $\mathbb{Z}$ which is not prime, and prove your claim.

Proposition 1.39. *Let R be a commutative ring with $1 \neq 0$. Then an ideal $P \subseteq R$ is prime if and only if R/P is an integral domain.*

Exercise 38. Prove the proposition above.

Corollary 1.40. *Every maximal ideal is prime.*

Proof. If $\mathfrak{m}$ is a maximal ideal, then $R/\mathfrak{m}$ is a field by Proposition 1.36, which is also an integral domain by a homework exercise, so $\mathfrak{m}$ is prime by Proposition 1.39. $\square$

Proposition 1.41. *Let R be a commutative ring with $1 \neq 0$, and let $x \in R$. Then x is not a unit if and only if there exists a maximal ideal $\mathfrak{m}$ with $x \in \mathfrak{m}$.*

Exercise 39. Prove the proposition above. You may freely use that every proper ideal is contained in some maximal ideal.

1.11 Localization and Rings of Fractions

Example 1.42. Note that we can define $\mathbb{Q}$ as

$$\mathbb{Q} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0 \right\} / \sim$$

where we say $\frac{a}{b} \sim \frac{c}{d}$ if $(ad - bc) = 0$.

More generally, we make the following definition:

Definition 1.43. Let R be a commutative ring, and $S \subseteq R$ be a *multiplicative set*, meaning that $1_R \in S$ and $a, b \in S \implies ab \in S$. Then we define $S^{-1}R$, called the **localization of R at S** by

$$S^{-1}R = \left\{ \frac{a}{b} \mid a \in R, b \in S \right\} / \sim$$

where we say $\frac{a}{b} \sim \frac{c}{d}$ if there exists an $s \in S$ such that $s(ad - bc) = 0$.

When S does not contain any zero-divisors, saying $s(ad - bc) = 0$ for some $s \in S$ is equivalent to saying $(ad - bc) = 0$. In this case, Dummit & Foote calls $S^{-1}R$ the **ring of fractions** of R with respect to S . Some authors use localization and ring of fractions interchangeably.

Exercise 40. Verify that $\sim$ is an equivalence relation.

Exercise 41. Explain how to view $\mathbb{Q}$ as a localization of some ring R at some multiplicative set S (i.e., what is R and what is S ?)

Proposition 1.44. Let R and S be as above. Then $S^{-1}R$ is a commutative ring with identity element $\frac{1_R}{1_R}$ and the operations:

$$\begin{aligned} \frac{a}{b} + \frac{c}{d} &= \frac{ad + bc}{bd} \\ \frac{a}{b} \cdot \frac{c}{d} &= \frac{ac}{bd} \end{aligned}$$

Proof. To see this, we should show that these are well-defined operations (meaning, we get the same answer regardless of which representatives we use), and further that they turn $S^{-1}R$ into a commutative ring, which we'll do in homework. $\square$

Definition 1.45. If R is an integral domain and $S = R \setminus \{0\}$, then $S^{-1}R$ is called the **quotient field** or **field of fractions** of R and is often denoted $Q(R)$. If R is not necessarily an integral domain, and S is the set of non-zero-divisors of R , then we call $S^{-1}R$ the **total quotient ring** or **total ring of fractions** of R and is still often denoted $Q(R)$.

Exercise 42. Explain why $Q(R)$ is a field when R is an integral domain, but will not be a field if R is not an integral domain.

Exercise 43. Let $\varphi : R \rightarrow S^{-1}R$ be the map defined by $\varphi(r) = r/1$. Describe $\ker \varphi$, and then prove that if S has no zero-divisors then φ is injective.

Proposition 1.46. Let R, T be commutative rings, and $\psi : R \rightarrow T$ a ring homomorphism. Let $S \subseteq R$ be a multiplicative set, and assume that $\psi(r)$ is invertible for every element $r \in S$. Then there exists a ring homomorphism $\pi : S^{-1}R \rightarrow T$ such that $\pi \circ \varphi = \psi$.

$$\begin{array}{ccc} R & \xrightarrow{\varphi} & S^{-1}R \\ & \searrow \psi & \vdots \exists \pi \\ & & T \end{array}$$

Remark 1.47. The proposition above is actually a **universal property** of the localization $S^{-1}R$. In category theory, we say that a property P is a universal property if whenever X and Y both satisfy P , then $X \cong Y$. In other words, universal properties uniquely identify objects up to isomorphism.

Exercise 44. Prove the proposition by defining π .

Example 1.48. Let R be a commutative ring, and let $f \in R$ be an element which is not nilpotent. Then $S := \{1, f, f^2, f^3, \dots\}$ is a multiplicative set, and $S^{-1}R = \{\frac{a}{f^n} \mid a \in R, n \geq 0\}$ is often denoted R_f , and is called R **localized at f** .

Exercise 45. Let R be a commutative ring and $f \in R[x]$. Use the first isomorphism theorem to prove that $R[x]/(xf - 1) \cong R_f$.

Example 1.49. Let R be a commutative ring, and let $P \subset R$ be a prime ideal. Then $S := R \setminus P$ is a multiplicative set, (we will check this) and $S^{-1}R = \{\frac{a}{b} \mid a \in R, b \notin P\}$ is often denoted R_P , and is called R **localized at P** .

Exercise 46. Check that if P is a prime ideal of a commutative ring R , then $R \setminus P$ is a multiplicative set, i.e. a multiplicatively closed set containing 1.

1.12 Products of Rings and the Chinese Remainder Theorem

Definition 1.50. Given a family of rings $\{R_i\}_{i \in \Lambda}$, we define the **product** of these rings by

$$\prod_{i \in \Lambda} R_i = \{(a_i)_{i \in \Lambda} \mid a_i \in R_i \text{ for } i \in \Lambda\}$$

with componentwise addition and multiplication, namely:

$$\begin{aligned} (a_i)_{i \in \Lambda} + (b_i)_{i \in \Lambda} &= (a_i + b_i)_{i \in \Lambda}, \text{ and} \\ (a_i)_{i \in \Lambda} \cdot (b_i)_{i \in \Lambda} &= (a_i b_i)_{i \in \Lambda} \end{aligned}$$

Note, when Λ is a finite set, we will often denote our elements of $\prod_{i \in \Lambda} R_i$ as finite tuples, for example $\prod_{i \in \{1,2,3\}} R_i = R_1 \times R_2 \times R_3 = \{(a_1, a_2, a_3) \mid a_i \in R_i\}$.

Exercise 47. Let $S = R_1 \times R_2$ for some rings R_1, R_2 . What is 0_S ? What about 1_S ?

Exercise 48. Find an integer x such that $x \equiv 4 \pmod{5}$ and $x \equiv 1 \pmod{6}$, and discuss your strategies with your group. Is it possible to find an integer x such that $x \equiv 1 \pmod{10}$ and $x \equiv 2 \pmod{6}$? Explain why or why not.

Theorem 1.51 (Chinese Remainder Theorem). *Let R be a commutative ring with $0 \neq 1$, and let $I_1, \dots, I_k$ be ideals of R . Let*

$$\varphi : R \rightarrow R/I_1 \times \cdots \times R/I_k$$

be defined by $\phi(x) = ([x]_{I_1}, \dots, [x]_{I_n}) = (x + I_1, \dots, x + I_n)$. Then

(a) φ is a ring homomorphism and $\ker \varphi = I_1 \cap \cdots \cap I_n$.

(b) If in addition $I_i + I_j = R$ for $i \neq j$ then φ is surjective and $I_1 \cap \cdots \cap I_n = I_1 \cdots I_n$. In particular,

$$R/(I_1 \cap \cdots \cap I_n) \cong R/I_1 \times \cdots \times R/I_n$$

Exercise 49. The proof that φ is a ring homomorphism is a direct check (you do not need to do this). Explain why $\ker \varphi = I_1 \cap \cdots \cap I_n$.

Exercise 50. Prove (b) for the case that $k = 2$: Let $\varphi : R \rightarrow R/I_2 \times R/I_2$ be the map defined above, and suppose that $I_1 + I_2 = R$, so that there is some $a \in I_1, b \in I_2$ such that $a + b = 1$. Prove that φ is surjective. You already showed $I_1 \cap I_2 = I_1 I_2$ in your last homework. (Hint: First compute $\varphi(a + b), \varphi(a)$, and $\varphi(b)$.)

Exercise 51. Use induction on k and the previous exercise to prove (b) in general.

Corollary 1.52. *Let $n \in \mathbb{Z}$ have a prime factorization $n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$, where p_i are distinct primes. Then $p_i^{\alpha_i} \mathbb{Z} + p_j^{\alpha_j} \mathbb{Z} = \mathbb{Z}$ for $i \neq j$ and so by the Chinese Remainder Theorem we have*

$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/p_1^{\alpha_1}\mathbb{Z} \times \cdots \times \mathbb{Z}/p_k^{\alpha_k}\mathbb{Z}$$

1.13 Euclidean Domains

Definition 1.53. Let R be an integral domain. A function $N : R \rightarrow \mathbb{Z}_{\geq 0}$ with $N(0) = 0$ is called a **norm** on R . If $N(a) > 0$ for $a \neq 0$, then N is called a **positive norm**.

Note that an integral domain R can have many different norms.

Definition 1.54. An integral domain R is called a **Euclidean domain** (or, possesses a division algorithm) if there exists a norm¹ N on R such that for any $a, b \in R$ with $b \neq 0$ there exists $q, r \in R$ such that

$$a = bq + r \text{ and } r = 0 \text{ or } N(r) < N(b)$$

Exercise 52. Define a norm on $\mathbb{Z}$ that shows $\mathbb{Z}$ is a Euclidean domain.

Exercise 53. Explain why if k is a field, then any norm on k shows it is a Euclidean domain.

Exercise 54. Let k be a field. Define a norm on $k[x]$ that shows $k[x]$ is a Euclidean domain.

Proposition 1.55. *If R is a Euclidean domain, then every ideal in R is principal (i.e. R is a PID). More precisely, if $(0) \neq I \subseteq R$ is an ideal, then $I = (\alpha)$ where α is any nonzero element of I of minimum norm.*

Proof. Let R be a Euclidean domain and $(0) \neq I \subseteq R$ an ideal. Let $\alpha \in I$ be a nonzero element of minimum norm, then clearly $(\alpha) \subseteq I$. Complete the proof by using the norm to show that $I \subseteq (\alpha)$.

□

¹If a norm satisfies these properties, Aluffi calls it a *Euclidean valuation*. However, *valuation* usually refers to a map satisfying other conditions, so we'll use the notation of Dummit&Foote and call this a Euclidean norm instead

Example 1.56. Consider the ring $\mathbb{Q}(\sqrt{d}) = \{a + b\sqrt{d} \mid a, b \in \mathbb{Q}\}$. The norm $N : \mathbb{Q}(\sqrt{d}) \rightarrow \mathbb{Z}_{\geq 0}$ given by $N(a + b\sqrt{d}) = a^2 - db^2$ is *multiplicative*, meaning that $N(ab) = N(a)N(b)$. Further, you showed in your homework that $\mathbb{Q}(\sqrt{d})$ is a field, and so is a Euclidean domain with this norm.

Example 1.57. Consider the ring $\mathbb{Z}[\sqrt{d}] = \{a + b\sqrt{d} \mid a, b \in \mathbb{Z}\}$, and define the norm $N : \mathbb{Z}[\sqrt{d}] \rightarrow \mathbb{Z}_{\geq 0}$ by $N(a + b\sqrt{d}) = a^2 - db^2$. Not all of the rings of the form $\mathbb{Z}[\sqrt{d}]$ will be Euclidean domains, and we will see why later.

When $d = 2, 3 \pmod{4}$, you can show that $\mathbb{Z}[\sqrt{d}]$ is the *integral closure* of $\mathbb{Z}$ in $\mathbb{Q}(\sqrt{d})$, which means $\mathbb{Z}[\sqrt{d}] = \{x \in \mathbb{Q}(\sqrt{d}) \mid x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0 \text{ for some } a_i \in \mathbb{Z}\}$. This means that $\mathbb{Z}[\sqrt{d}]$ is the *ring of integers* of $\mathbb{Q}(\sqrt{d})$ when $d = 2, 3 \pmod{4}$.

Example 1.58. When $d = -1$, we get ring $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$, with norm $N : \mathbb{Z}[i] \rightarrow \mathbb{Z}_{\geq 0}$ by $N(a + b\sqrt{d}) = a^2 + b^2$, which is in fact a Euclidean domain (with this norm).

Definition 1.59. Let R be a commutative ring with $a, b \in R$ and $b \neq 0$. Then

1. We say a is a **multiple** of b (or b divides a , denoted $b \mid a$) if there exists an $x \in R$ such that $a = bx$.
2. We say d is a **greatest common divisor** of a and b (denoted $d = \gcd(a, b)$) if $d \mid a$, $d \mid b$ and for any d' dividing both a and b we have $d' \mid d$.

In general, a gcd of two elements may not exist.

Exercise 55. Let R be an integral domain and let $a, b \in R$, and suppose that d, d' are both gcds of a and b . Then

1. $d = d'u$ for some unit $u \in R$, and
2. $(d) = (d')$.

Proposition 1.60. *Let R be a Euclidean domain, and let $a, b \in R \setminus \{0\}$. Suppose we run the Euclidean algorithm on a and b , and $d = r_n$ is the last nonzero remainder, namely:*

$$\begin{aligned} a &= q_0b + r_0 \\ b &= q_1r_0 + r_1 \\ r_0 &= q_2r_1 + r_2 \\ &\vdots \\ r_{n-2} &= q_nr_{n-1} + r_n \\ r_{n-1} &= q_{n+1}r_n + 0. \end{aligned}$$

Then $d = \gcd(a, b)$ and $(a, b) = (d)$.

Exercise 56. Prove the Proposition as follows:

1. First, explain why the Euclidean algorithm terminates, using the definition of Euclidean domain.
2. Show that $d \mid r_i$ for all i and that $d \mid a$ and $d \mid b$, and conclude that $(d) \supseteq (a, b)$.
3. Write d as a linear combination of a and b , and conclude that $(d) \subseteq (a, b)$.
4. Prove that if $d' \mid a$ and $d' \mid b$, then $d' \mid d$ (in other words, $d = \gcd(a, b)$).

Remark 1.61. We've shown above that in a Euclidean domain, every two elements have a gcd, which we can find using the Euclidean algorithm.

1.14 Principal Ideal Domains (PIDs)

Recall that a **principal ideal domain** is an integral domain in which every ideal is principal, or generated by one element.

Remark 1.62. We have already proved that every Euclidean domain is a PID in Proposition 1.55, but in general, not every PID is a Euclidean domain. For example, it can be shown that the ring $\mathbb{Z}\left[\frac{1+\sqrt{19}}{2}\right] = \left\{a + b\left(\frac{1+\sqrt{19}}{2}\right) \mid a, b \in \mathbb{Z}\right\}$ is a PID but not a Euclidean domain (See Dummit & Foote, Section 8.2).

Proposition 1.63. *Let R be an integral domain and a, b be nonzero elements of R . Assume there is some $d \in R$ such that $(d) = (a, b)$. Then*

(a) $d = \gcd(a, b)$

(b) d can be written as $d = ra + sb$ for some $r, s \in R$, and

(c) d is unique up to multiplication by units.

Exercise 57. Prove the proposition above.

Example 1.64. Note that in a PID, we always have a d such that $(d) = (a, b)$, but in general we may not have such a d . We've seen $\mathbb{Z}[x]$ is not a PID, and $\gcd(2, x) = 1$ but $(1) \neq (2, x)$.

Proposition 1.65. *Let R be a PID and $(0) \neq P \subseteq R$ be a prime ideal. Then P is maximal.*

Exercise 58. Prove the proposition above. Note that in a general commutative ring, we have shown that maximal ideals are prime, so this Proposition shows that in a PID, P is prime if and only if P is maximal.

1.15 Unique Factorization Domains (UFDs)

In this section, we'll define unique factorization domains which have unique (up to units) factorizations into irreducible. Before we define a UFD, we will need to define irreducible.

Definition 1.66. Let R be an integral domain, and let $0 \neq r, s \in R$ be nonzero elements in R which are not units.

1. We say r is **irreducible** if whenever $r = ab$, then either a or b is a unit in R .
2. We say r is **prime** if (r) is a prime ideal, i.e. if whenever $r \mid ab$ we have $r \mid a$ or $r \mid b$.
3. We say r, s are **associates** if $r = us$ for some unit u (or equivalently, if $(r) = (s)$).

Exercise 59. Let R be an integral domain. Prove that every prime element is irreducible.

Exercise 60. Let R be a PID. Prove that an element is prime element if and only if it is irreducible.

Exercise 61. Prove that 3 is irreducible in $\mathbb{Z}[\sqrt{-5}]$ but is not prime (Hint: Note that $3^2 = 9 = (2 + \sqrt{-5})(2 - \sqrt{-5})$).

Definition 1.67. Let R be an integral domain. Then R is called a **unique factorization domain (UFD)** if for every $0 \neq r \in R$, which is not a unit, we can write

$$r = p_1 \cdots p_n$$

for some irreducible elements p_i , and this decomposition is unique up to associates and permutation. (In other words, if $r = q_1 \cdots q_m$ for some other irreducibles q_i , then $m = n$ and for each i , $p_i = uq_j$ for some unit u and some j .)

Example 1.68. Consider the following:

1. Any field is a UFD (there are no nonzero elements which are not units).
2. $\mathbb{Z}$ is a UFD (and PID). We have unique factorization into primes.
3. $\mathbb{Z}[\sqrt{-5}]$ is not a UFD, since $6 = 2 \cdot 3 = (1 - \sqrt{-5})(1 + \sqrt{-5})$.

Theorem 1.69. *Let R be a PID. Then R is a UFD.*

Exercise 62. Prove the theorem above using the following outline:

1. Fix a nonzero non unit $a \in R$. Prove that a is a *finite* product of irreducible elements. If a is not irreducible, then we can write $a = bc$ for some b, c . If this process doesn't terminate in a finite product of irreducibles, form a chain of ideals to get a contradiction.
2. Now if $a = p_1 \cdots p_n = q_1 \cdots q_m$ are two factorizations into irreducibles, then use the fact that each p_i and q_i is prime to show $n = m$ and each p_i is an associate to some q_j .

Proposition 1.70. *If R is a UFD, then $0 \neq p \in R$ is prime if and only if p is irreducible.*

Proof. We showed in Exercise 59 that if p is prime, then p is irreducible. Now suppose p is irreducible, and suppose $ab \in (p)$. Then $ab = rp$ for some $r \in R$, and we factor r, a and b into irreducibles, namely $r = p_1 \cdots p_n$, and $a = q_1 \cdots q_k$, $b = q_{k+1} \cdots q_m$. Then we have that

$$(p_1 \cdots p_n)p = rp = ab = q_1 \cdots q_m$$

are two different factorizations of ab into irreducibles, which means that p is an associate of some q_i , so that $p \mid a$ or $p \mid b$, i.e. $a \in (p)$ or $b \in (p)$. $\square$

Remark 1.71. We have shown the following, but the reverse implications do not hold:

$$\text{Euclidean domain} \implies \text{PID} \implies \text{UFD}$$

1.16 The ring $\mathbb{Z}[i]$

Consider the ring $R = \mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$, which is a Euclidean domain with the norm $N(a + bi) = a^2 + b^2$. In the next few exercises, we will investigate properties of this ring.

Exercise 63. Find every invertible element of $\mathbb{Z}[i]$. (Hint: Use the fact that the norm is multiplicative).

Exercise 64. Show that if $\alpha \in \mathbb{Z}[i]$ is prime, $\alpha \mid p$ for some prime $p \in \mathbb{Z}$. (Hint: If $\alpha = a + bi$, consider the prime factorization of $(a + bi)(a - bi)$).

Exercise 65. Show that if $N(\alpha)$ is a prime number in $\mathbb{Z}$, then α is a prime element in $\mathbb{Z}[i]$. Then, check that $1 + i$ and $1 - i$ are prime in $\mathbb{Z}[i]$.

Exercise 66. Show that if p is prime in $\mathbb{Z}$ and $p \equiv 3 \pmod{4}$, then p is prime in $\mathbb{Z}[i]$.

Exercise 67. Show that if p is prime in $\mathbb{Z}$ and $p \equiv 1 \pmod{4}$, then $p = \alpha \bar{\alpha}$ for some prime element $\alpha \in \mathbb{Z}[i]$, where $\bar{\alpha}$ here denotes the complex conjugate of α . (Hint: First show $p \mid x^2 + 1$ where $x = (p - 1)!$, and then show p is not prime in $\mathbb{Z}[i]$)

Exercise 68. Use the previous exercises to characterize all prime elements of $\mathbb{Z}[i]$.

1.17 Polynomial UFDs

In this section, we will prove the following theorem:

Theorem 1.72. *If R is a UFD, then $R[x]$ is a UFD.*

Exercise 69. Explain why the theorem above tells us that if R is a UFD, then $R[x_1, \dots, x_n]$ is a UFD.

Proposition 1.73. *Let $I \subseteq R$ be an ideal of a UFD R , and let $IR[x] = I[x] = \{a_n x^n + \dots + a_1 x + a_0 \mid n \in \mathbb{N}, a_i \in I\}$. Then $IR[x]$ is an ideal of $R[x]$ and*

$$\frac{R[x]}{IR[x]} \cong (R/I)[x]$$

Exercise 70. Prove the proposition above using the First Isomorphism Theorem. Note that we know the kernel of any ring homomorphism is an ideal (of the domain), so if you show that $IR[x]$ is the kernel of some ring homomorphism $\varphi : R[x] \rightarrow (R/I)[x]$, then you do not need to explicitly check that it is an ideal.

Remark 1.74. A corollary of this Proposition is that for any $a \in R$, we have $R[x]/(x-a) \cong R$, which you already proved directly in Homework 2.

Theorem 1.75. *Let k be a field. Then $k[x]$ is a Euclidean domain with respect to the norm $N : k[x] \rightarrow \mathbb{Z}_{\geq 0}$ given by $N(f) = \deg(f)$. (Note: Then we also know that $k[x]$ is a PID and UFD)*

Notation 1.76. Let R be an integral domain. Recall that we call non-units $r, s \in R \setminus \{0\}$ **associates** if $r = us$ for some unit $u \in R$. In this case, we write $r \sim s$.

Definition 1.77. Let R be a UFD, and $f = a_0 + a_1x + \cdots + a_nx^n \in R[x]$. Then we define the **content of f** , by $c(f) := \gcd(a_0, \dots, a_n) \in R$. We say f is **primitive** if $c(f) \sim 1_R$.

Exercise 71. Determine $c(f)$ for $f = 2x^2 + 10x + 4 \in \mathbb{Z}[x]$ and then give an example of $g \in \mathbb{Z}[x]$ such that g is primitive.

Lemma 1.78 (Gauss' Lemma). *Let R be a UFD and $f, g \in R[x]$. Then*

$$c(fg) \sim c(f)c(g)$$

Proof. Note that we can write $f = c(f)f_1$ and $g = c(g)g_1$ where f_1, g_1 are primitive. Thus, $fg = c(f)c(g)f_1g_1$. It suffices to show that f_1g_1 is primitive, and we will proceed by contradiction. Suppose

$$\begin{aligned} f_1 &= a_0 + a_1x + \cdots + a_nx^n \\ g_1 &= b_0 + b_1x + \cdots + b_mx^m \end{aligned}$$

and suppose f_1g_1 is not primitive, so let $p \mid c(f_1g_1)$ be a prime dividing all coefficients of f_1g_1 . Since f_1 and g_1 are primitive, we know $p \nmid c(f_1)$ and $p \nmid c(g_1)$, so we can let

$$\begin{aligned} \ell &= \min\{i \mid p \nmid a_i\}, \text{ and} \\ k &= \min\{i \mid p \nmid b_i\} \end{aligned}$$

Complete the proof by finding a contradiction.

□

Lemma 1.79. *Let R be a UFD, let $K = Q(R)$ be its quotient field, and let $f, g \in R[x]$ with f primitive. Then $f \mid g$ in $R[x]$ if and only if $f \mid g$ in $K[x]$.*

Exercise 72. Prove the lemma above. For the $(\leftarrow)$ direction, write the definition of $f \mid g$ in $K[x]$, and note that if $h \in K[x]$, you can multiply by some $a \in R$ to “clear denominators” and get $ah \in R[x]$. Then use Gauss’ Lemma.

Lemma 1.80. *Let R be a UFD and $K = Q(R)$, and suppose $f \in R[x]$ is primitive. Then f is prime in $R[x]$ if and only if f is prime in $K[X]$.*

Exercise 73. Prove the Lemma above.

Lemma 1.81. *Let R be an integral domain, and assume that every element of R is a finite product of prime elements. Then*

- (a) *If x is an irreducible element of R , then x is prime.*
- (b) *R is a UFD.*

Proof. Recall that in an integral domain, we showed all prime elements are irreducible. Then if x is irreducible and $x = p_1 \dots p_n$ is a product of primes, we must have that $n = 1$ and $x = p_1$, proving (a).

Since prime elements are irreducible, we have that every element of R has a factorization into irreducibles, so it suffices to show this factorization is unique (up to ordering/associates). Let $r \neq 0$ be a non-unit, and suppose $r = p_1 \dots p_n = q_1 \dots q_m$ are two factorizations into primes (which are irreducibles). Then since p_i is prime and $p_i \mid q_1 \dots q_m$, we have that $p_i \mid q_j$ for some j , and since q_j is prime $p_i \sim q_j$. Continuing as such, we see that $m = n$ and each p_i is an associate of some q_j . $\square$

Theorem 1.82. *Let R be a UFD. Then $R[x]$ is a UFD.*

Exercise 74. Let $f \in R[x] \setminus \{0\}$ be a non-unit. We will show it factors into primes (and so also irreducibles) by induction on the degree of f . First prove the base case $\deg(f) = 0$.

Exercise 75. Assume that every polynomial of degree $< m$ has a unique factorization into primes, and let $f \in R[x]$ have degree m . Show that there is a prime $p \in R[x]$ such that $p \mid f$, and $\deg(p) > 0$. (Hint: Let $K = Q(R)$, and consider what happens in $K[X]$).

Exercise 76. Complete the proof of Theorem 1.82

Example 1.83. Note that this theorem tells us that $\mathbb{Z}[x]$ is a UFD, but we know that $\mathbb{Z}[x]$ is not a PID, since $(2, x)$ is not principal.

Remark 1.84. The proof above will not work for infinite power series. In particular, if R is a UFD, we do not necessarily have that $R[[x]]$ is a UFD. However, it can be shown that if R is a PID, then $R[[x]]$ is a UFD, and further, if k is a field, then $k[[x]]$ is a Euclidean Domain.

1.18 Irreducibility Criteria

Proposition 1.85. *Let k be a field, $f \in k[x]$, $\alpha \in k$. Then $(x - \alpha) \mid f$ if and only if $f(\alpha) = 0$.*

Exercise 77. Prove the proposition above.

Note that you have already proved this (in fact, for $R[x]$ for any commutative ring) directly on Homework 1 by noting that if $f = \sum_{i=0}^n a_i x^i$, then $f(x) - f(\alpha) = \sum_{i=0}^n a_i (x^i - \alpha^i)$ and thus $(x - \alpha) \mid (f(x) - f(\alpha))$ since $(x - \alpha)$ divides every term $(x^i - \alpha^i)$, so that if $f(\alpha) = 0$, then $(x - \alpha) \mid f(x)$.

Now, however, in the case that k is a field, try to find a quicker proof using the fact that $k[x]$ is a Euclidean domain with the norm $N(f) = \deg f$.

Corollary 1.86. *Let k be a field, and $f \in k[x]$, such that $\deg f = 2$ or 3 . Then f is irreducible in $k[x]$ if and only if f has no roots in k .*

Proposition 1.87. *Let $f = a_n x^n + \cdots + a_1 x + a_0 \in \mathbb{Z}[x]$ with $a_n \neq 0$. Let $\alpha = \frac{r}{s} \in \mathbb{Q}$ (where $r, s \in \mathbb{Z}$ and $\gcd(r, s) = 1$). Then $r \mid a_0$ and $s \mid a_n$.*

Exercise 78. Prove the proposition above by plugging $\frac{r}{s}$ into f and clearing denominators.

Exercise 79. Explain why $x^3 - 3x - 1$ is irreducible in $\mathbb{Q}[x]$.

Definition 1.88. We say a polynomial $f \in R[x]$ is **monic** if its leading coefficient is 1, i.e. if $f = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$ for some $a_i \in R$.

Exercise 80. Let R be an integral domain, and let $f \in R[x]$ be monic and $I \subseteq R$ be an ideal. Assume $\bar{f} \in (R/I)[x]$ cannot be factored into polynomials of smaller degree. Prove by contradiction that f is irreducible in $R[x]$.

Exercise 81. Let $f = x^2 + x + 1 \in \mathbb{Z}[x]$, $I = 2\mathbb{Z}$. Show that $\bar{f} = x^2 + x + \bar{1} \in \mathbb{Z}/(2\mathbb{Z})[x]$ is irreducible in $\mathbb{Z}/(2\mathbb{Z})[x]$ (check roots) and conclude that f is irreducible in $\mathbb{Z}[x]$.

Exercise 82. Let S be an integral domain, and $g, h \in S[x]$, with $\deg g, h \geq 1$. Prove that if $x^n = gh$, then $x|h$ and $x|g$. (Note that if S is not a domain, this is not true. For example, in $\mathbb{Z}/4\mathbb{Z}$, we have $x^2 = (2x + 1)(2x^3 + x^2)$).

Proposition 1.89 (Eisenstein's Criterion). *Let R be an integral domain, and let $P \subseteq R$ be a prime ideal. Let*

$$f = x^n + a_{n-1}x^{n-1} + \dots + a_0 \in R[x]$$

with $n \geq 1$. Assume that $a_{n-1}, a_{n-2}, \dots, a_1, a_0 \in P$ and $a_0 \in P^2$. Then f is irreducible in $R[x]$.

Exercise 83. Prove the proposition above. Assume for contradiction that $f = gh$ and then use Exercises 80 and 82 with $S = R/P$, and consider the constant terms of $\bar{g}, \bar{h}$.

Proposition 1.90 (Eisenstein's Criterion). *Let R be a UFD, $K = Q(R)$, let*

$$f = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 \in R[x]$$

with $n \geq 1$. Assume that there is a prime element $p \in R$ such that $p \nmid a_n$, $p \mid a_i$ for $0 \leq i \leq n-1$ and $p^2 \nmid a_0$. Then f is irreducible in $K[x]$. (If f is primitive this is equivalent to f is irreducible in $R[x]$.)

Proof. Let $f = c(f)f_1$ where $f_1 \in R[x]$ is primitive. Since $p \nmid a_n$, we have that $p \nmid c(f)$, so that f_1 satisfies the same conditions as f . Thus, we may assume f is primitive.

Now assume $f = gh$, where $g = b_m x^m + \cdots + b_0$ and $h = c_\ell x^\ell + \cdots + c_0$. Since $p \mid a_0 = b_0 c_0$, we have that $p \mid b_0$ or $p \mid c_0$. Without loss of generality assume $p \mid b_0$. Since $p \nmid a_n$, we know $p \nmid b_m c_\ell$, so $p \nmid b_m$. Let $k = \min\{j \mid p \text{ does not divide } b_j\}$, and note that $1 \leq k \leq m < n$. Note that

$$a_k = b_0 c_k + b_1 c_{k-1} + \cdots + b_k c_0 \equiv b_k c_0 \pmod{p}$$

Since $k < n$, we have $p \mid a_k$, so $p \mid b_k c_0$, but $p \nmid b_k$ (by choice of k), so we must have that $p \mid c_0$. Thus, $p^2 \mid b_0 c_0$, giving us a contradiction. $\square$

Exercise 84. Check that the following are irreducible, using Eisenstein's criterion:

1. $x^4 + 20x + 5 \in \mathbb{Z}[x]$.
2. $z^4 + y^2 x z^2 + x^2 y z + x y^3 \in k[x, y, z]$, where k is a field (Think about $k[x, y, x]$ as $R[z]$ where $R = k[x, y]$).

Remark 1.91. Let k be a field, and let $f \in k[x] \setminus \{0\}$. Then the following are equivalent:

- (a) f is irreducible
- (b) f is prime
- (c) $(f) \subseteq k[x]$ is a prime ideal
- (d) (f) is a maximal ideal
- (e) $k[x]/I$ is a field.

Exercise 85. Explain why the remark above is true (note that we have proved many directions already!)

Proposition 1.92. *Let $f \in k[x]$ where k is a field, and suppose $\deg f \geq 1$. Let $f = f_1^{n_1} \cdots f_k^{n_k}$ be a factorization of f into irreducibles, which are pairwise not associate. Then*

$$\frac{k[x]}{f} \cong \frac{k[x]}{f_1^{n_1}} \times \cdots \times \frac{k[x]}{f_k^{n_k}}$$

Exercise 86. Prove the proposition using the Chinese Remainder Theorem.

Definition 1.93. Let R be an integral domain, let $f \in R[x]$, and $c \in R$. We say c is a root of f of **multiplicity** m if $f = (x - c)^m g$ for some $g \in R[x]$ such that $(x - c) \nmid g$ (or equivalently, $(x - c)^m \mid f$ but $(x - c)^{m+1} \nmid f$)

Proposition 1.94. *Let K be a field. If $a_1, \dots, a_k$ are distinct roots of $f \in K[x]$ with multiplicities $m_1, \dots, m_k$ respectively, then*

$$(x - a_1)^{m_1} (x - a_2)^{m_2} \cdots (x - a_k)^{m_k} \mid f$$

Proof. Note that $(x - a_i)$ are irreducible and for $i \neq j$ are not associate, so that since each of these divide f their product must. □